Term Project - CS329E

CS329 E - Elements of Data Analytics
(10 points)

1 Description

The objective of this project assignment is to create a comprehensive data analytics project utilizing a real-world dataset. Your primary responsibilities will include selecting a suitable public dataset, cleaning and preparing the data, and applying one or more data analytic models from the course to address your research questions.

Your project can focus on various data analytics models such as regression, classification, clustering, or any other relevant techniques. Once you have chosen a model or combination of models, you will implement the selected data model(s) using the scikit-learn library or other appropriate tools, evaluate the effectiveness of the model(s), and provide a thorough interpretation of your findings.

To successfully complete this assignment, it is crucial to document each step of the data analysis process and to provide clear and insightful explanations of your project results. This will not only showcase your technical skills in data analytics but also demonstrate your ability to communicate complex data-driven insights to a broader audience.

In your project, you should use one of the models or techniques that we have not covered in our class. You should do your research and learn them on your own. Here is a list of Models or Topics that we have not covered yet or will not cover in our class.

- Cross Decomposition
- Feature Selection for Supervised Learning
- Multiclass and multioutput algorithms
- Data Clustering with MeanShift
- Data Clustering with Birch
- Data Clustering with Gaussian Mixture Models
- Data Clustering with Spectral Clustering
- Topic modelling Latent Dirichlet Allocation (LDA)
- Outlier Detection Techniques, One-Class SVM
- Techniques to deal with Unbalanced Datasets
- Nonlinear Dimensionality Reduction Technique t-SNE

2 Public Dataset List

First select a data set from the available public datasets.

You can find a list of such public data sets here:
http://www.teymourian.de/public-data-sets-for-data-analytic-projects/)
Some of the important data collection links are:

- Machine Learning Data sets https://archive.ics.uci.edu/
- https://www.data.gov/
- https://www.kaggle.com/datasets

Notes:

- The data set must not be very Large.
- You can reuse one of the assignment data sets.

3 Define a Project (2 Points)

Define a data science research question based on the selected dataset. You can select one of the listed datasets or search on the web, and pick up one of the available public datasets.

You should describe the following items

1. Describe well the new model that you are using.
2. What is your data set about?
3. Clean up your data and reduce it if your dataset is very large.
4. What is exactly your research question? What do you want to learn from data? What is your learning model ,e.g., a Classification, Clustering, etc?
5. What is your current expectations about the results? Why?
6. Describe how you evaluate your project? How to access the correctness of your model? How well would you expect that the model will work?

4 Implementation (6 Points)

- You need to implement your project in python. You are allowed to use any Machine Learning Library like scikit-learn[1]. You are also allowed to implement your project without using any libraries.
- You can use Jupyter Notebooks to implement your project and provide your documentations.
- You code should be completed and be compilable without any errors. We should be able to read your documents, and be able to run your project.

[1] https://scikit-learn.org/

- Run your implementation (on your Laptop or on Cluster if data is large) and generate the results.
- Provide the interpretations of your results.
- What can you do to improve your results? Apply your ideas to improve your results.
- Provide any references that you have used.

5 Create Recorded Presentation Video of your Project (2 Points)

- Create a document or a presentation (You can create a presentation using your Jupyter notebook, or create a powerpoint presentation, or other formats) to describe your project and results
- Describe well the new Model or Technique!
- Describe your project and your code.
- Describe the results of your project in a professional way.
- You may want to visualization diagrams and describe the results based on some diagrams - but having diagrams is not a MUST have to get the full credit.
- Describe the model and results of your project in a way that every person in this field can read, enjoy and understand.

Record your presentation in form of a video recording. Present your talk on your laptop computer and record it using a desktop recording software.

You should use your webcam so that we can identify who is presenter.

You have the following 2 options to create a presentation video:

- **Method 1:** You can record your presentation using Zoom client application, start your zoom client, start a zoom meeting (you will be the only participant), click on share your desktop, turn on camera and record your presentation, end the zoom session and you will get a MP4 video file of your presentation. Upload the file to Blackboard system
- **Method 2:** Alternatively, you can use any desktop recording software, generate a MP4 file and upload MP4 to Canvas, or upload it to some other cloud storage (like google Drive, DropBox) and share with us the URL Link of your presentation (submit the link in Canvas). Make sure that you give use access to your file. Make sure that we can access your file.

Note 1: Your video presentation should be between 8 minutes to maximum 15 minutes long (Min 8 mi, Max 15 min)

Note 3: You should turn on your Web Camera so that we can see your face during the presentation, and be able to identify the presenter.

We will review your video and upload it later into the Class Video Gallery for Class Members to review.

Grading will be based on quality of your presentation, and correct describing of algorithms or concepts.

5.1 Turnins

- Submit your Jupyter Notebook or Python code
- Submit your documentation (It can be part of your Jupyter Notebook or a separate file like PPT or Docs, or other formats)
- Submit your Video Presentation. Upload your video file to Canvas Assignment. We will later download it and upload to Panopto Video Gallery for the entire class.

5.2 Grading Criteria

Your term project will be graded based on the following rubrics:

1. Correct description of the selected dataset.
2. Correctness of your data preparation and cleanup approach.
3. General Correctness of data analysis approach.
4. Correctness of your implementation and results.
5. Correctness and completeness of the interpretations of your results.
6. completeness of all provided references including all code and idea sources.